

ASSIGNMENT-2

Create a Vehicle class with different data members that have public or private access modifiers. Define getters and setters for the private members. Define default and parameterized constructors. Define member methods to start, stop, accelerate vehicle and to calculate mileage.

Print the Vehicle details in a tabular format to compare the mileage of vehicles with different fuel types.

CODE:

Vehicle.java

```
public class Vehicle {  
    public String brandname;  
    public String modelname;  
    public java.time.Year yearofmfg;  
    public String color;  
    public char fueltype; // p=petrol,e=electric,d=diesel,c=cng  
    public float mileage;  
    public float price;  
    public int seats;  
    private String mfgcode;  
    private int noofservices;  
  
    public void setMfgCode(String mCode){  
        mfgcode=mCode;  
    }  
    public String getMfgCode(){
```

```
        return mfgcode;
    }

    public void setnoofservices(int mCode){
        noofservices=mCode;
    }

    public int getnoofservices(){
        return noofservices;
    }

    public Vehicle(){
        brandname = "MG";
        modelname = "Hector";
        yearofmfg = java.time.Year.of(2025);
        color = "Red";
        fueltype = 'p';
        seats = 5;
        price = 115000.45f;
    }

    public Vehicle(String brandname,String modelName,float price,String color){
        this.brandname=brandname;
        this.modelname=modelName;
        this.price=price;
        this.color=color;
    }

    public Vehicle(char fueltype,float price,String mfgcode){
        this.fueltype=fueltype;
        this.price=price;
        this.mfgcode=mfgcode;
    }
}
```

```
}  
  
public void start() {  
    System.out.println("Start Ignition by pressing the button");  
    System.out.println("Your initial Speed is 10kmph");  
}
```

```
  
public void drive() {  
    System.out.println("Let's Go!");  
}
```

```
  
public int accelerate(int initSp) {  
    return initSp + 20;  
}
```

```
  
public void stop() {  
    System.out.println("Stop Ignition by pressing the button");  
    System.out.println("Your Vehicle is stopped!");  
}
```

```
  
public float calcMileage(float fuelAmt, float distance) {  
    return distance / fuelAmt;  
}
```

```
  
/*public static void main(String[] args) {  
    Vehicle v = new Vehicle();  
    v.brandname = "MG";
```

```

        v.modelname = "Hector";

        v.yearofmfg = java.time.Year.of(2025);

        v.color = "Red";

        v.fueltype = 'p';

        v.seats = 5;

        v.price = 115000.45f;

        printDetails(v);

        v.start();

        v.drive();


        int newSp = v.accelarate(40);

        System.out.println("New Speed: " + newSp + " kmph");


        float mileage = v.calcMileage(40, 500.2f);

        System.out.println("The mileage is: " + mileage + " kmpL");

        v.stop();

    }*/

}

```

MainForVehicle.java

```

public class MainForVehicle {

    public static void main(String[] a) {

        Vehicle v = new Vehicle();

        v.brandname = "MG";
    }
}

```

```
v.modelname = "Hector";  
v.yearofmfg = java.time.Year.of(2025);  
v.color = "Red";  
v.fueltype = 'p';  
v.seats = 5;  
v.price = 115000.45f;
```

```
Vehicle v1 = new Vehicle("Toyota", "Canry", 250000.99f, "Red");  
v1.setMfgCode("MAh93e4");  
v1.seats = 5;  
Vehicle v2 = new Vehicle('D', 2000053.35f, "Mah68740");
```

```
v.start();
```

```
v.drive();
```

```
Vehicle[] fleet = new Vehicle[] { v, v1, v2 };
```

```
System.out.println("Brand\tModel\tYear\tColor\tFuel\tSeats\tPrice\tMileage\tMfgCode\tService  
s");
```

```
for (Vehicle v32 : fleet) {
```

```
    v32.start();
```

```
    v32.drive();
```

```
    v32.accelerate(20);
```

```
    if (v32.fueltype == 'D') {
```

```
        float m322 = v32.calcMileage(50, 500);
```

```
        printTabular(v32, m322 );
```

```
    } else if (v32.fueltype == 'P' || v32.fueltype == 'C') {
```

```
        float m322 = v32.calcMileage(40, 500);
```

```

        printTabular(v32, m322 );
    } else {
        float m322 = 0;
        printTabular(v32, m322 );
    }

}

}

```

```

public static void printTabular(Vehicle v, float m) {

    System.out.println(v.brandname + "\t" + v.modelname + "\t" + v.yearofmfg + "\t" + v.color + "\t" +
v.fueltype
        + "\t" + v.seats + "\t" + v.price + "\t" + m + "\t" + v.getMfgCode() + "\t" + v.getnoofservices());
}

}

```

OUTPUT:

```

Start Ignition by pressing the button
Your initial Speed is 10kmph
Let's Go!
Brand  Model  Year  Color  Fuel  Seats  Price  Mileage  MfgCode  Services
Start Ignition by pressing the button
Your initial Speed is 10kmph
Let's Go!
MG      Hector  2025   Red    p      5      115000.45      0.0      null      0
Start Ignition by pressing the button
Your initial Speed is 10kmph
Let's Go!
Toyota  Canry   null   Red      5      250000.98      0.0      MAh93e4  0
Start Ignition by pressing the button
Your initial Speed is 10kmph
Let's Go!
null    null    null    null    D      0      2000053.4      10.0      Mah68740  0
PS C:\Users\Kunal_jhindal\OneDrive\Desktop\Sem4\PJL\Assignment 2>

```